

Testing, safely rolling out and monitoring ML pricing

Extra baggage and extra-legroom seats | Current Pricing vs New Model | Business Rules keep final control

Etihad already prices ancillaries in-house. The job is not to build a model that prints a price. It is to test a **New Model** against **Current Pricing**, keep **Business Rules** in control, roll out in traffic steps, and HOLD or **Return to Current Pricing** when the model is not healthy.

Current Pricing	New Model
Simulated existing system using route, days to departure, product, channel, and remaining inventory.	P(buy price, booking context), then Price x P(buy) on an allowed grid. Trained on Databricks; scored in the lab from an exported artifact.

Business Rules always finish the job (included-in-fare, loyalty, min/max, inventory, airport vs online bags, temporary RM caps). Example: recommend AED 175, cap AED 150, customer sees AED 150.

Products, rollout, checkout

v1 products: extra baggage and preferred / extra-legroom seat. Fare-brand upgrades are future expansion, not built here.

Rollout: Offline to shadow (log New Model, customer still gets Current Pricing) to 5% to 20% to 50% to 100%. Pause or Return to Current Pricing at any time.

Checkout fallback: New Model to Current Pricing to simple rules to a safe fixed price. If the model fails at checkout, the customer still gets a valid price.

How to read the four screens

1. Price Explanation	Why this booking's extra-legroom price moved versus Current Pricing. Both prices in AED.
2. Pricing Controls	Traffic percent, price bands, temporary caps. Not per-passenger approval.
3. Pricing Test Results	Revenue per passenger, conversion, average selling price. Simulated comparison.
4. Production Monitor	Commercial, model, system, and rules health, then a Rollout Decision: HOLD / expand / return.

Two-minute walk

Price Explanation	Show why the New Model recommends a different extra-legroom price than Current Pricing on AUH-LHR Economy Basic, four days out. Both prices are in AED.
Pricing Controls	Revenue Management sets traffic, bands, pause, or Return to Current Pricing. Not per-passenger approval.
Pricing Test Results	Ask whether revenue per passenger rose without damaging conversion. Say out loud: simulated result using synthetic/public data.
Production Monitor	If behaviour drifts or quality drops, HOLD expansion or Return to Current Pricing. Checkout still has a fallback.

Resume bullets

Built a production lab that tests a New Model against Current Pricing for extra baggage and extra-legroom seats, then applies Business Rules so Revenue Management keeps final control.

Trained P(buy | price, booking context) on Databricks, selected price by expected revenue, and compared Current vs New on simulated offer logs (synthetic + public data).

Added rollout (shadow to 5% to 20%), checkout fallback, and a monitor with a HOLD / Return to Current Pricing decision so a model is not left live when behaviour drifts.

Honesty line for every interview answer: Simulated result using synthetic/public data. Do not claim Etihad employment, Etihad data, or a live airline A/B test.